Status	Finished
Started	Sunday, 18 October 2026, 9:18 AM
Completed	Sunday, 18 October 2026, 9:22 AM
Duration	3 mins 35 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching

b. Merging

c. Sorting

d. Complexity

The correct answer is:
Complexity

Question **2**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,80,90,150,200,300,400,470,70,120

b. 80,150,200,470,40,90,300,400,70,120

c. 200,470,80,150,40,90,300,400,70,120

d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **3**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Rearranging
- c. Merging
- d. Searching

The correct answer is: Sorting

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes data harder to manage
- b. Provides a basis for other algorithms to work efficiently
- c. Decreases the efficiency of algorithms
- d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **5**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Insertion sort
- b. Bubble sort
- c. Quick sort
- d. Heap sort

The correct answer is:
Insertion sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Insertion Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Binary Search
c. Linear Search
d. Bubble Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It is based on the divide-and-conquer approach
c. It is less efficient than bubble sort
d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **9**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is less efficient for large lists
- b. Bubble Sort is difficult to implement
- c. Bubble Sort does not guarantee sorted order
- d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **10**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By sorting data in a single pass
- b. By comparing elements sequentially
- c. By breaking the input into smaller parts and merging them
- d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **11**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Deleting data from a list
- b. Inserting data into a list
- c. Arranging data in a particular format
- d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **12**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data sequentially
b. Rearranging data without sorting
c. Dividing the input into parts, solving each part, and combining the solutions
d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting makes it harder to search for items
c. Sorting helps in finding duplicates quickly
d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **14**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Bubbling up the largest element to its correct position with each pass
c. Sorting data in a non-sequential manner
d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **15**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists
b. It bubbles up the largest element in each pass
c. It uses the heap data structure
d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass